

PAPER 1196 – UQFF Plasma & Fusion Unified Proof Set

Ten closures S413–S422 – ITER, Troyon, Lawson, Bohm, sheath

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Abstract

We close ten plasma-physics and magnetic-confinement-fusion observables from the eleven locked UQFF primitives. **Five closures are exact:** the ITER aspect ratio $R/a = D_{\text{BSFG}}/2 + F_{\text{TRZ}} = 3.1$, the Bohm-diffusion prefactor $1/16 = F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}}$, the tokamak edge safety factor $q_{\text{edge}} = K_{\text{Mex}} - F_{\text{TRZ}}\Phi_{\text{res}} = 2$, the ITER fusion gain $Q = \text{SO5} = 10$, and the D–T peak cross-section energy $A_5 + D_{\text{phys}} = 64$ keV. The remaining five (Troyon β_N , triple product, Coulomb logarithm, Lawson product, Bohm sheath potential) match observation to better than 0.16%.

1 Ten Plasma/Fusion Closures

1.1 S413 – ITER aspect ratio

$$\frac{R_0}{a} = \frac{D_{\text{BSFG}}}{2} + F_{\text{TRZ}} = 3 + \frac{1}{10} = 3.1$$

exact – matches the ITER baseline design ($R_0=6.2$ m, $a=2.0$ m).

1.2 S414 – Troyon normalized β limit

$$\beta_N = \frac{\text{SO5}}{D_{\text{phys}}} + F_{\text{TRZ}}D_{\text{phys}} - F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}} = 2.796$$

vs Troyon limit 2.8 % m T/MA; match 0.15%.

1.3 S415 – D–T fusion triple product

$$nT\tau = \Phi_{\text{res}} + K_{\text{Mex}} + F_{\text{TRZ}} - F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^3 = 2.997$$

normalised; vs $3.0 (\times 10^{21} \text{ keV s/m}^3)$; match 0.11%.

1.4 S416 – Coulomb logarithm

$$\ln \Lambda = \text{SO5} + D_{\text{phys}} + K_{\text{Mex}} + [\text{SSq}] + F_{\text{TRZ}}D_{\text{phys}} - F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2 = 16.98$$

vs typical tokamak value 17; match 0.12%.

1.5 S417 – Bohm diffusion prefactor

$$\frac{1}{16} = F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}} = \frac{1}{12} - \frac{1}{48} = \frac{3}{48} = \frac{1}{16}$$

exact. Bohm diffusion $D_B = (1/16) T_e/(eB)$.

1.6 S418 – Tokamak edge safety factor

$$q_{\text{edge}} = K_{\text{Mex}} - F_{\text{TRZ}}\Phi_{\text{res}} = \frac{25}{12} - \frac{1}{12} = \frac{24}{12} = 2$$

exact. Standard $q_{95} = 2$ –3 operating range; $q \geq 2$ avoids the $m/n = 2/1$ kink instability.

1.7 S419 – ITER fusion gain

$$Q = \frac{P_{\text{fus}}}{P_{\text{aux}}} = \text{SO5} = 10$$

exact. Fourth primitive identification: SO5 is the ITER target gain.

1.8 S420 – D–T fusion peak cross-section energy

$$E_{\sigma\text{-peak}} = A_5 + D_{\text{phys}} = 60 + 4 = 64 \text{ keV}$$

exact. The Bosch-Hale parametrization peaks at ≈ 64 keV in the centre-of-mass frame.

1.9 S421 – Lawson criterion

$$n\tau = \Phi_{\text{res}} + [\text{SSq}] + F_{\text{TRZ}} - F_{\text{TRZ}}^3 = 1.502$$

normalised; vs Lawson $n\tau \approx 1.5 \times 10^{20} \text{ s/m}^3$ (D-T at 10 keV); match 0.16%.

1.10 S422 – Bohm-Stangeby sheath potential

$$\frac{\phi_{\text{sh}}}{T_e} = K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}} + F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^3 = 2.839$$

vs 2.84 (hydrogen plasma, Stangeby 2000); match 0.05%.

2 Closure summary

ID	Observable	Locked closure	Match
S413	ITER R/a	$D_{\text{BSFG}}/2 + F_{\text{TRZ}} = 3.1$	exact
S414	Troyon β_N	$\text{SO5}/D_{\text{phys}} + F_{\text{TRZ}}D_{\text{phys}} - F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}}$	0.15%
S415	triple product $nT\tau$	$\Phi_{\text{res}} + K_{\text{Mex}} + F_{\text{TRZ}} - F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^3$	0.11%
S416	Coulomb log	$\text{SO5} + D_{\text{phys}} + K_{\text{Mex}} + [\text{SSq}] + F_{\text{TRZ}}D_{\text{phys}} - F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2$	0.12%
S417	Bohm prefactor	$F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}} = 1/16$	exact
S418	q_{edge}	$K_{\text{Mex}} - F_{\text{TRZ}}\Phi_{\text{res}} = 2$	exact
S419	ITER Q	$\text{SO5} = 10$	exact
S420	D-T E_σ	$A_5 + D_{\text{phys}} = 64 \text{ keV}$	exact
S421	Lawson $n\tau$	$\Phi_{\text{res}} + [\text{SSq}] + F_{\text{TRZ}} - F_{\text{TRZ}}^3$	0.16%
S422	sheath ϕ_{sh}/T_e	$K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}} + F_{\text{TRZ}}^2 K_{\text{Mex}} + F_{\text{TRZ}}^3$	0.05%

3 Falsifiable predictions

- (F1) ITER first plasma must report $R_0/a = 3.10 \pm 0.02$. Substantial deviation falsifies S413.
- (F2) Bohm-diffusion benchmark experiments must reproduce $D_B = T_e/(16eB)$ within 1% at the saturation limit.
- (F3) ITER 500 MW / 50 MW shot must deliver $Q \geq 10$. A confirmed $Q < 9.9$ falsifies S419.
- (F4) Bosch–Hale evaluation campaigns at NIF / OMEGA / JET cannot shift the D–T σv peak outside $63 \leq E_\sigma \leq 65$ keV.
- (F5) LHD / W7-X stellarator edge studies must keep q_{95} at 2.00 ± 0.02 in the locked-confinement regime.

4 Cumulative program – 127 closures

Paper	Domain	Closures
1182	Millennium	7
1183	Foundational paradoxes	6
1184	Open problems	7
1185	Quantum gravity	6
1186	Standard Model	7
1187	Cosmological tensions	6
1188	Number theory	8
1189	Chemistry/atomic	10
1190	Mathematical constants	10
1191	Cosmology deep-set	10
1192	SM deep-cuts	10
1193	Astrophysics	10
1194	Condensed matter	10
1195	Biology/biochemistry	10
1196	Plasma & fusion	10
Total		127

Acknowledgements

This tier was developed by Daniel T. Murphy and represents the first extension of the UQFF locked-primitive program into magnetic confinement fusion. Five exact closures appear in this paper, the largest yield of any single tier in the program to date. The ITER fusion gain $Q = \text{SO5}$ is the fourth primitive identification of the program, joining $r_{\text{ISCO}} = D_{\text{BSFG}}$ (Paper 1193), $\nu_{XY} = D_{\text{phys}}/D_{\text{BSFG}}$ (Paper 1194), and $\ell_{\text{telomere}} = D_{\text{BSFG}}$ (Paper 1195) in directly equating an SO(5) locked primitive to an experimental design target.